

Topics on Robotics & Vision

M.Sc Autonomous Mobile Robotics

Introduction to the Lecture

LectureSlide

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1. Introduction

Introduction



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Content Preview



- The goal of this lecture is to give you a better view of how mobile robotics works, the different challenges it faces and methods to overcome them.
- This lecture you will use ROS, a versatile software capable of doing pathfinding and localisation, a perfect fit for use in mobile robotics.
- This lecture is a total of 3 SWS with a total of forty-five (45) UE.
- A unit (UE) is defined as 45 min lecture.



- Lecture materials and all possible supplements will be present in its Github Repo.
 - You can easily access the link to the web-page from [here](#).

Github is chosen for easy access to material management and CI/CD capabilities and allowing hosting websites.

- In the lecture content is also distributed as a WebBook which can be accessed from the [Repo website](#).



- The student should be comfortable with working with either Python and/or C++ and should have an beginner level knowledge of the Linux environment and the command line interface

Requirements	Taught Lecture	Code	Degree	Outcome
C/C++ Programming	Programming II	PRO2	B.Sc	ROS 2 Programming
Python Programming	Software Design	SWD	B.Sc	Linux Experience
-	-	IOT	B.Sc	Programming
-	-	-	-	-
-	-	-	-	-

Table 1: Distribution of materials across the semester.



Description	Value
Official Name	Autonome mobile Robotik
Lecture Code	AMR
Module Code	MECH-M-DUAL-3-AMR
Lecture Name	Autonomous Mobile Robotics
Semester	3
Season	WS
Lecturer	Daniel T. McGuinness, Ph.D
Module Responsible	BnM interim
Software	Linux, ROS, Python
SWS Total	3
UE Total	45
ECTS	5
Work Language	English



- The lecture will have two personal assignments which will be based on Mobile robotics, ROS and programming applications related to it. More details will be given in the assignment.

Assignment Type	Value
Personal Assignment A	50
Personal Assignment B	50
Sum	100



Title
A very informal journey through ROS 2
A Concise Introduction to Robot Programming with ROS2
Programming Principles and Practice using C++
Introduction to Autonomous Mobile Robots

Table 2: Lecture sources which can be useful during the course of the lecture. For more information on sources, please consult the [repo](#).



Topic	Units	Self Study
Locomotion	4	8
Mobile Robot Kinematics	4	8
Perception	4	8
Mobile Robot Localisation	4	8
Welcome to Linux	4	8
Command Line Fundamentals	4	8
Working with Files and Folders	4	8
Installation	1	8
ROS Concepts	4	8



Topic	Units	Self Study
Command Line Tools	4	8
Client Libraries	4	8
Transform Library	4	8
SUM	45	90